Module 5: Analysis

Analyzer Introduction

The last of the (main) projects…

1. Intermediate Representation

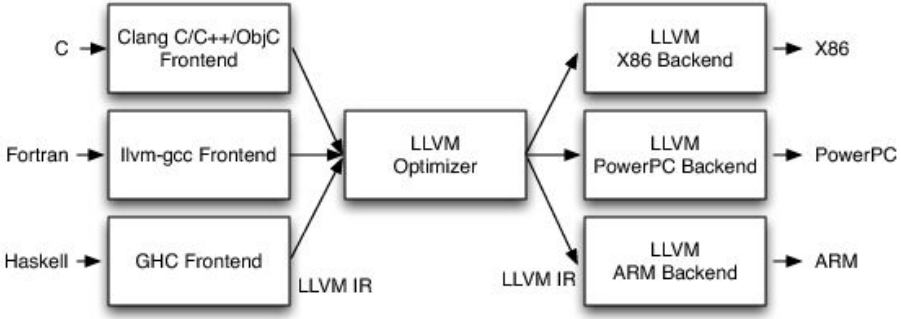
Despite the length, it's actually pretty simple!

Intermediate Representation (IR)

Definitions

An **IR** represents code *somewhere* between the AST and final output.

- Crudely, AST with type information.
- Typically, the IR is the first step of resolving references and types.
  - Later IR steps may simplify structure or optimize code.
- Each “phase” of compilation typically involves a different IR.



Recall our “Big 3” analysis goals:

1. Enforce additional AST requirements
2. Resolve types, variables, functions, etc.
3. Perform type checking of values

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